

Sam Crouch

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EDUCATION

Missouri University of Science and Technology

Rolla, MO

B.S. Computer Science

May 2027

Relevant Coursework: Data Structures, Algorithms, Operating Systems, Programming Languages, Microcontrollers and Embedded Systems, File Structures and Database Systems, Introduction to Artificial Intelligence

EXPERIENCE

Controls Intern

May 2026 – Aug. 2026

Altec

Saint Joseph, MO

- Developed a Qt/QML library for embedded Linux displays, standardizing UI components across multiple product lines
- Designed and Implemented a C++ daemon to expose hardware functionality through a text file interface, decoupling application logic from low-level hardware control

Forklift Operator

May 2025 – Aug. 2025

Menards

Saint Peters, MO

- Safely operated forklifts in a lumber yard and warehouse
- Maintained product inventory and order fulfillment

Lube Tech

Nov. 2022 – Aug. 2024

Valvoline Instant Oil Change

Saint Peters, MO

- Performed preventative maintenance services including oil changes, fluid replacements, and filter installations
- Conducted vehicle inspections and communicated maintenance recommendations to customers

PROJECTS

pyVPW | *Python*

Sep. 2025 – Present

- Developed an open-source Python library implementing the SAE J1850 VPW protocol for GM vehicle diagnostics
- Designed the API to simplify development of diagnostic and data-logging tools
- Implemented the GM seed-key security algorithm to unlock PCMs, enabling writes to protected memory for tuning and diagnostic applications
- Published the project on GitHub with documentation and example applications, supporting adoption by other developers

LLM-TUI | *Python, Textual*

Feb. 2026 – May 2026

- Built a customizable, terminal-based AI assistant using Python and the Textual framework
- Designed an asynchronous terminal UI supporting real-time streaming responses, on-the-fly model switching, file attachments, and a snipping-tool style screenshot tool
- Integrated multiple LLM providers including OpenAI, Anthropic, Deepseek, Grok, and Ollama behind a common interface, enabling seamless provider switching

TECHNICAL SKILLS

Languages: C, C++, Python, QML, Bash, SQL, HTML/CSS

Frameworks: Qt, Textual, Django

Developer Tools: Git, Docker, CMake, Conan, Linux, GDB

Embedded/Automotive: Embedded Linux, CAN Bus, UART, AVR Microcontrollers